

Case Study: VAEP implementation on FIFA World Cup 2022 Dataset

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1. Introduction

VAEP (Valuing Actions by Estimating Probabilities) is a data-driven sports analytics framework that aims to measure the objective contribution of each on-the-ball action a soccer player takes for a match. The VAEP models soccer as a series of game states, whereas traditional statistics only account for the finishing touch (such as goals or assists). Using a computation of the change in probability of a team scoring or conceding goals in the next k actions ($k = 10$), VAEP assigns a net value (in units of expected goal difference) to every pass, dribble, clearance, tackle, interception and shot. **Why VAEP?**

Holistic Action Valuation: Match events are more often than not non-shooting actions, the majority of events are holistic actions (pass, take-on, ball recovery, clearance etc.). The conventional statistics such as Goals, Assists or Expected Goals (xG) do not appreciate how important the build up is to a chance being created or how crucial the defensive play is to being denied a chance.

Context-Aware Performance: A pass in the attacking third under pressure is worth far more than a good safe pass in the base centre-backs zone, as it is assessed based on the context in which it is made and not by itself.

Decoupled Offensive & Defensive Evaluation: VAEP breaks out each action's worth into an offensive contribution (ΔP_{scores}) and a defensive contribution ($-\Delta P_{\text{concedes}}$), giving a multi-dimensional profile of each player's performance, rather than just a number.

Temporal Lookahead: It takes into account the longer-term ripple effect of an action (as opposed to the immediate effect) – such as a line-breaking pass that creates space for a goal several actions later.

2. Dataset

2.1. Train-Test Dataset Strategy

Training Set

When training, the pipeline dynamically fetches matches from the live StatsBomb open-data catalogue for men's international tournaments (World Cup, UEFA Euro, Copa América, Africa Cup of Nations); excludes the 2022 World Cup (as it is excluded from the catalogue for now); uses the 5 most recent tournaments that are eligible (i.e. ones that are currently included in the catalogue) to train.

In this run, that automatic selection resolved to:

Copa América 2024 (32 matches) UEFA Euro 2024 (51 matches) Africa Cup of Nations (AFCON) 2023 (52 matches) UEFA Euro 2020 (51 matches) FIFA World Cup 2018 (64 matches)

ORCID(s):

Total Training Scale: 250 matches, converted into SPADL action sequences and then rolled into overlapping 3-action game states for feature extraction.

2.1.1. Test Set

The tournament FIFA World Cup 2022 (64 matches) was deliberately left out of the training query and strictly limited to the model inference, evaluation and player rating process as the benchmark tournament. It is a tournament level (temporal) split and not a random split: the model is not exposed to a single action from the 2022 World Cup during training, which makes it a more challenging split of generalization than a random split

2.2. Dataset Preprocessing

StatsBomb's StatsBombLoader exposes hundreds of raw event types, nested JSON metadata, and vendor-specific tagging (duel pairs, off-ball markers, referee/half-time events) that are not directly usable as ML features. Only active, on-the-ball player actions were parsed via `socceraction.spadl.statsbomb.convert_to_actions` and mapped into uniform structures. Games with missing or corrupted event data are skipped individually (with a console warning) rather than failing the whole pipeline — this robustness fix matters because open StatsBomb data occasionally has gaps for newer tournaments.

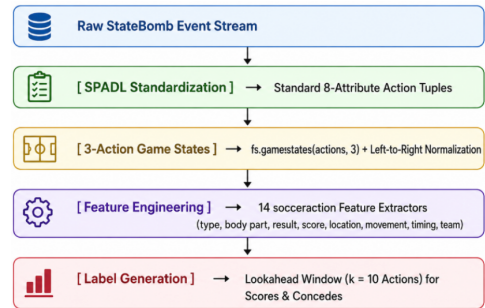


Fig. 1. Dataset preprocessing flowchart.

2.3. SPADL

SPADL stands for Soccer Player Action Description Language. SPADL transforms event formats from many different sources into a common language that can be understood by humans. Each action is represented in a predefined 9-attribute tuple:

Action $a_i = \langle \text{StartTime}, \text{EndTime}, \text{StartLoc}, \text{EndLoc}, \text{Player}, \text{Team}, \text{ActionType}, \text{BodyPart}, \text{Result} \rangle$

Standardized Action Types: The action types observed in this pipeline include pass, cross, dribble,

take_on, shot, shot_penalty, shot_freekick, corner_crossed, corner_short, throw_in, clearance, tackle, interception, foul, bad_touch, keeper_save, keeper_claim, keeper_punch, and keeper_pick_up.

Coordinate Normalization: Coordinates are normalized to standardised pitch dimensions. Each action is also reoriented from left to right relative to the home team's attacking direction using `fs.play_left_to_right(gamestates, home_team_id)`. This makes positional features comparable regardless of the half or side from which a team started.

2.4. Pipeline

2.4.1. Game State Representation:

Each action a_i is contextualized using a rolling window of the last 3 actions, $S_i = [a_{i-2}, a_{i-1}, a_i]$, built with `fs.gamestates(actions, 3)`.

2.4.2. Feature Extractor Functions(14 generators from `socceraction.vaep.features`):

- **SPADL Primitive Features:** `actiontype`, `actiontype_onehot`, `bodypart`, `bodypart_onehot`, `result`, and `result_onehot` — the raw category label plus a one-hot encoding for each.
- **Spatial Features:** `startlocation`, `endlocation`, `movement` (distance/direction covered during the action), and `space_delta` (spatial change relative to the previous action in the game state).
- **Temporal & Game-Context Features:** `time_delta` (time between consecutive actions), `time` (elapsed match time), `goalscore` (current score state for each side), and `team` (possession/team-change indicator).

2.4.3. Data Preprocessing & Robustness Fixes:

- **Dynamic ID Fetching:** The competition/season IDs are queried live via `SBL.competitions()` and filtered by gender and competition name, instead of being hard-coded — this avoids stale-ID 404 errors if StatsBomb's catalogue changes.
- **Categorical/NaN Fix:** Pandas category-typed columns are cast back to object before `.fillna(0)` is applied, which avoids a `TypeError` that otherwise occurs when trying to fill missing values in a strictly-typed categorical column — this was a required fix for feeding data into the neural network.
- **Boolean → Numeric Casting:** All boolean indicator columns (True/False) are explicitly cast to int (1/0), and any remaining non-numeric/text columns are stripped out via `select_dtypes(exclude=['object', 'category'])` before being passed to any of the five models, ensuring mathematical compatibility across both gradient-boosting and neural architectures.
- **Minimum Game Length Filter:** Any match with fewer than 10 total actions is skipped, since it cannot support a full $k = 10$ lookahead label.

3. Models

Let game state $S_i = [a_{i-2}, a_{i-1}, a_i]$ represent the match state after action a_i . Let $P_{scores}(S_i, x)$ and $P_{concedes}(S_i, x)$ denote the probability that team x (currently in possession) scores or concedes a goal within the next $k = 10$ actions:

Offensive Value:

$$\Delta P_{scores}(a_i, x) = P_{scores}(S_i, x) - P_{scores}(S_{i-1}, x)$$

Defensive Value:

$$\begin{aligned} -\Delta P_{concedes}(a_i, x) &= -\left(P_{concedes}(S_i, x) - P_{concedes}(S_{i-1}, x) \right) \\ &= P_{concedes}(S_{i-1}, x) - P_{concedes}(S_i, x) \end{aligned}$$

Total VAEP Value:

$$V(a_i, x) = \Delta P_{scores}(a_i, x) + \left(-\Delta P_{concedes}(a_i, x) \right)$$

$$PlayerRating(p) = \sum_{a_i \in A_p} V(a_i).$$

This is the summed contribution across all of a player's actions in the tournament.

This project implements this formula computation using `socceraction.vaep.formula.value(A_test, p_scores_series, p_concedes_series)`, which takes the two learned probability estimates and returns the offensive value, defensive value, and total VAEP value for every action in the test set.

Implemented Model Architectures

This iteration of the project trains and compares five independent classifier families, each fit twice (once for scores, once for concedes), rather than the three originally planned. Text and one-hot features are stripped before training (`select_dtypes(exclude=['object', 'category'])`); Logistic Regression additionally requires `StandardScaler`-normalized inputs since it is sensitive to feature scale, while the tree-based models and MLP are trained on the unscaled numeric feature matrix.

Baseline: CatBoost Classifier

Architecture: Symmetric (oblivious) decision trees, `iterations=300`, `depth=6`, `learning_rate=0.08`.

Loss Function: Binary cross-entropy (log loss):

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{j=1}^N \left[y_j \log(p_j) + (1 - y_j) \log(1 - p_j) \right].$$

Why Used: This is the reference architecture taken from the VAEP paper. Oblivious trees offer a quick CPU inference, free from overfitting, and excellent support for high-cardinality 1-hot categorical features.

Logistic Regression (Linear Baseline)

Architecture: LogisticRegression.
Configured with `max_iter=1000`.
Trained on StandardScaler-normalized numeric features.

Mathematical Formulation:

A linear decision boundary passed through a sigmoid link:

$$P(y = 1 | x) = \sigma(w^T x + b),$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Why Used: Is a sanity-check baseline in that if the much more complicated tree ensembles and neural network cannot meaningfully outperform the linear model, then it would be a sign of model complexity being doing the work, and not engineered features. The non-linear models do outperform it – as illustrated below – but by a modest amount.

Calibrated LightGBM (Histogram Gradient Boosting)

Architecture: Leaf-wise (best-first) tree growth.
`n_estimators=200` and `max_depth=5`.
`learning_rate=0.05`.
Wrapped in CalibratedClassifierCV.
Configured with `method='isotonic'` and `cv=3`.

Probability Calibration (Isotonic Regression):

Minimizes non-parametric squared error to make raw tree scores match true empirical probabilities:

$$\min_{\hat{m}} \sum_j (y_j - \hat{m}(f_j))^2,$$

subject to $\hat{m}(f_a) \leq \hat{m}(f_b)$
whenever $f_a \leq f_b$.

Why Used: We can compare the two, because the performance of LightGBM is not due to the leaf-wise approach to histogram building, but due to exact/approximate greedy split finding, so it is informative to compare the two to determine if LightGBM's strong performance is due to gradient boosting in general on this feature set, or due to this particular implementation. The results below demonstrate that XGBoost scores AUCs that closely match LightGBM's scores AUCs and is better than LightGBM on Concedes AUC.

Deep MLP Classifier (Neural Network)

Architecture: MLPClassifier with
`hidden_layer_sizes=(128, 64)`.
`activation='relu'` and `max_iter=20`.
A fully connected feed-forward network (input $\rightarrow 128 \rightarrow 64 \rightarrow 1$).

Mathematical Layer Formulation:

$$h^{(1)} = ReLU(W_1 x + b_1),$$

$$h^{(2)} = ReLU(W_2 h^{(1)} + b_2),$$

$$P = \sigma(W_3 h^{(2)} + b_3).$$

where $ReLU(z) = \max(0, z)$ and $\sigma(z) = \frac{1}{1 + e^{-z}}$.

Optimization: Adam optimizer with backpropagation and mini-batch gradient descent, minimizing cross-entropy loss, capped at 20 iterations.

XGBoost Classifier

Architecture: XGBClassifier with `n_estimators=200`, `max_depth=5`, and `learning_rate=0.05` — a second, independently-implemented gradient-boosting library used as a cross-check against LightGBM's histogram-based boosting.

Why Used: XGBoost uses exact/approximate greedy split-finding rather than LightGBM's leaf-wise histogram approach, so comparing the two indicates whether LightGBM's strong performance is a property of gradient boosting in general on this feature set, or an artifact of one specific implementation. As the results below show, XGBoost matches LightGBM's Scores AUC almost exactly and edges it out on Concedes AUC.

4. Model Evaluation

All five models were evaluated on the completely unseen 2022 World Cup test set using ROC-AUC (discrimination ability) as the headline metric:

Model	Scores AUC	Concedes AUC
CatBoost (Baseline)	0.8173	0.8361
Calibrated LightGBM	0.8202	0.8115
XGBoost	0.8202	0.8207
Logistic Regression	0.8040	0.8189
Deep MLP	0.8027	0.7580

Table 1

ROC-AUC scores for all five models on the 2022 World Cup test set.

The chart below shows that CatBoost, LightGBM, and XGBoost had comparable scores and the lowest (best) Brier score loss on both tasks, MLP was slightly worse-calibrated (Brier score loss is lower the better), and Logistic Regression was significantly worse-calibrated (noticeably the highest bar on the Scores-Brier panel) and comparable to its lower AUC.

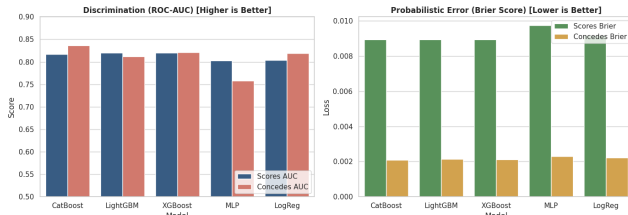


Fig. 2. AUC and Brier-score comparison across the five models.

All three tree-based ensembles (CatBoost, LightGBM, XGBoost) are in a narrow range around 0.80–0.84 AUC, which definitely outperform the MLP’s Concedes AUC (0.758, the lowest on the board). It is reasonable to expect that the right panel would be about 4× larger than the left because actions are far more likely to be labelled “scores” than “concedes” is likely to be immediately preceded by an action, so the problem of estimating the action is more difficult and noisier when the action is “scores”.

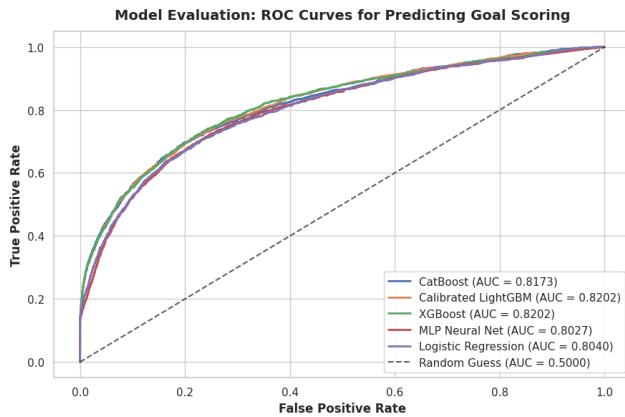


Fig. 3. ROC curves for the scoring and conceding models.

Early separation of the five curves from the diagonal “random guess” curve, which is the characteristic of a model that can distinguish the small number of “meaningful” actions (shots, take-ons into the box, through-balls) that increase the probability of scoring from the large number of “neutral” actions (actions that do not increase the probability of a goal). The curves of CatBoost, Calibrated LightGBM and XGBoost are nearly identical and are slightly higher than Logistic regression and the MLP on most of the range, as is confirmed in the AUC table above.

Why Tree Models Outperform the Deep MLP on This Data

- Extreme Class Imbalance:** This is a rare event classification problem, as only a small fraction of actions results directly in a goal or a concession within the 10-action window. The tree-based algorithms divide the feature space into rectangular partitions, with the boundaries being aligned to the axes, thus creating a partition structure that is better suited to the isolation of rare “leads to goal” branches than a smooth decision surface that is given by a neural network.

- LightGBM Early-Stopping Behavior:** In the training phase, LightGBM prints messages like “No further splits with positive gain, best gain: -inf”, which means that the tree is being trained on uninformative branches, starving itself of unpredictable data from the majority class (no-goal) and reducing the risk of overfitting on the noise.
- MLP Limitations on One-Hot Tabular Data:** To construct smooth linear hyperplanes, ReLU layers are used in the MLP. The one-hot categorical variables (sparse) with the spatial/temporal continuous variables tend to cause neural gradients to focus on the majority negative class, which aligns with the MLP in this class having the lowest Concedes AUC (0.758) of all five models.
- XGBoost vs. LightGBM Parity:** Two completely different gradient-boosting libraries achieve nearly the same Scores AUC (0.8202 for both of them), so this performance mark is a good benchmark of actual, extractable signal in the engineered features, rather than of some particular optimizer within one library.

5. Results & Analysis

Comparison Among Action Types (Goals/Shots vs. Passes vs. Defending)

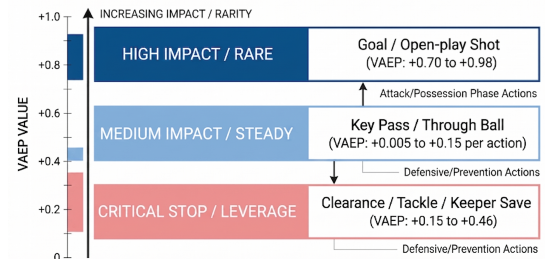


Fig. 4. VAEP action-value classification.

Goals & Shots: High peak values, typically +0.70 to +0.98 VAEP. As seen in the top-actions table below, successful open-play shots (e.g., Achraf Dari +0.980, Gonçalo Ramos +0.969) represent immediate state transitions where scoring probability surges from a low baseline toward 1.0 in a single action.

Passes & Assists: Passes individually yield smaller incremental values (roughly +0.005 to +0.15 in most cases), but elite playmakers accumulate large tournament-level VAEP totals through sheer volume of progressive, high-completion passing into dangerous zones — this is exactly the dynamic visualized in the Traditional-Metrics-vs-VAEP scatter plot below.

Tackles & Interceptions: Defensive actions derive their value primarily from reducing danger ($-\Delta P_{concedes}$). An open-field tackle or a last-man clearance inside the penalty box (e.g., Abdessamad Ezzalzouli’s clearance at +0.462,

Rank	Player	Action Type	Receiver	VAEP Value	Offensive Value	Defensive Value
1	Achraf Dari	shot	N/A	+0.980	+0.980	-0.000
2	Gonalo Matias Ramos	shot	N/A	+0.969	+0.968	+0.001
3	Daizen Maeda	shot	N/A	+0.963	+0.962	+0.001
4	Duřan Vlahovi	shot	N/A	+0.963	+0.963	-0.000
5	Young-Gwon Kim	shot	N/A	+0.963	+0.959	+0.003
6	Osman Bukari	shot	N/A	+0.960	+0.957	+0.003
7	Kalidou Koulibaly	shot	N/A	+0.960	+0.959	+0.001
8	Jean-Charles Castelletto	shot	N/A	+0.959	+0.959	+0.000
9	Boulaye Dia	shot	N/A	+0.957	+0.955	+0.001
10	Bukayo Saka	shot	N/A	+0.955	+0.954	+0.000

Table 2
Top 10 actions by raw VAEP value.

Rank	Player	Action Type	Defensive Value (ΔP_{def})	Result	Match Context
1	Abdessamad Ezzalzouli	clearance	+0.462	success	High-pressure box clearance
2	Damián Emiliano Mart�nez	keeper_save	+0.392	success	Crucial 1-on-1 goal-preventing save
3	Abdessamad Ezzalzouli	interception	+0.338	success	Cut off high-threat transition
4	�scar Duarte	tackle	+0.280	success	Last-ditch penalty box tackle
5	Young-Gwon Kim	tackle	+0.245	success	Vital stop in central box
6	Seyed Majid Hosseini	tackle	+0.230	success	Open-field transition tackle
7	Ibrahima Konat�	interception	+0.229	fail	High-leverage deflection disruption
8	Bryan Oviedo	tackle	+0.208	success	Wing-containment sliding tackle
9	Joseff Morrell	tackle	+0.190	fail	Tactical break-up foul/tackle
10	Seung-Gyu Kim	keeper_save	+0.190	fail	High-probability shot parry

Table 3
Top 10 defensive actions by defensive value (ΔP_{def}).

Emiliano Mart nez’s save at +0.392) delivers large single-action defensive value by preventing an opponent’s high-probability scoring opportunity.

Top 10 Actions in FIFA World Cup 2022 (Non-Penalty)

(Passes and crosses in this ranking would also carry a receiver or the next player to touch the ball in the same match, but the absolute top 10 by raw VAEP value are all successful shots, since a shot is the single action type capable of moving scoring probability all the way to ~ 1.0 in one step.)

Top 10 Defensive Actions in FIFA World Cup 2022

Table 3 shows us the top 10 defensive actions rated by our VAEP model:

Traditional Metrics (Goals + Assists) vs. Total VAEP

To test whether traditional box-score output fully captures a player’s contribution, Goals (successful shots) and Assists (the action immediately preceding a goal, in the same match and same team, from a pass/cross/take-on type) were combined into a G+A statistic and plotted against Total VAEP for every player with more than 100 actions in the tournament.

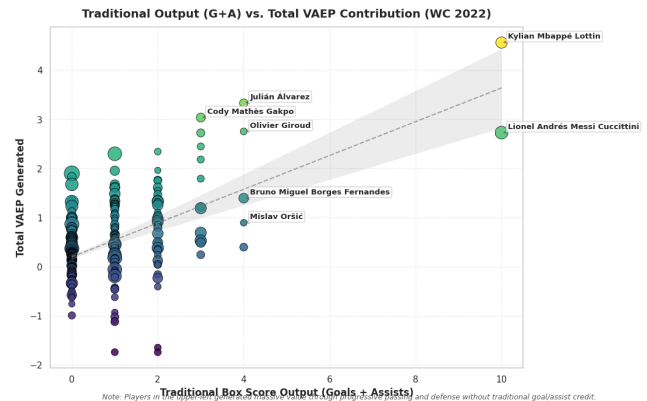


Fig. 5. Goals and assists versus total VAEP.

Both Mbapp  and Messi are in the leftmost corner ($G+A = 10$), as the two most prolific traditional players in the tournament, and their raw VAEP numbers are the highest too ($G+A = 10$). The more interesting signal is in the middle of the chart, those three guys: Julian  lvarez, Cody  akpo and Olivier Giroud have their Total VAEP nicely above the dashed trendline, which means that their $G+A$ total is producing more value than it would normally do, as they generated additional value through movement, progressive passing and defensive contribution. On the other hand, Bruno Fernandes and Mislav Or i did not sit up to the trendline — while they played in a similar style of play

Rank	Player Name	Team	Total VAEP	Offensive VAEP	Defensive VAEP	Action Count
1	Kylian Mbappé	France	4.562	5.965	-1.403	658
2	Julián Álvarez	Argentina	3.330	3.474	-0.144	293
3	Cody Gakpo	Netherlands	3.035	3.067	-0.032	353
4	Olivier Giroud	France	2.753	2.829	-0.076	151
5	Lionel Messi	Argentina	2.728	5.119	-2.391	873
6	Bukayo Saka	England	2.721	2.736	-0.015	265
7	Richarlison	Brazil	2.446	2.506	-0.060	169
8	Gonçalo Ramos	Portugal	2.399	2.490	-0.091	84
9	Ritsu Doan	Japan	2.342	2.359	-0.018	150
10	Joško Gvardiol	Croatia	2.298	1.487	+0.811	1033

Table 4

Top 10 players by Total VAEP, FIFA World Cup 2022.

as some of the higher VAEP players they had a similar G+A figure, their total value creation from the play over the complete 90 minutes was less, indicating a greater dependency on the final scoring/assisting play than the sustained delivery throughout the game. This is the exact area where a player can tally the same number of goals and assists and make different kinds of contributions to a team's attack-defense cycle, which is what VAEP is meant to surface.

5. Top 10 Players in FIFA World Cup 2022

Overall Player Summary Table

The grouped bar chart makes the offense/defense trade-off visible at a glance. Every player except Gvardiol shows a negative (orange) Defensive VAEP bar — for attacking players this is expected and not a criticism: it reflects that turnovers deep in the opposition half slightly raise the counter-attacking team's chance of scoring against them. Messi and Mbappé have by far the largest negative defensive bars (-2.391 and -1.403), which is the direct cost of being the primary ball-carriers who most often lose possession in advanced, high-risk areas. Gvardiol is the visual outlier of the whole chart: he is the only player with a positive (rightward) orange bar, immediately marking him as a defender whose net contribution comes from a completely different source than everyone else on the list.

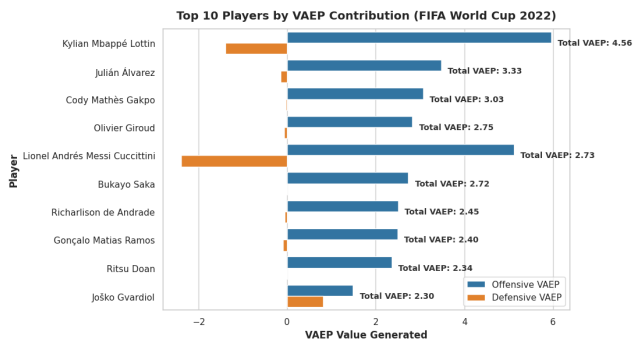


Fig. 6. Offensive, defensive, and total VAEP for the top 10 players.

This ranking can easily be justified, which is given below:

- Kylian Mbappé:** Mbappé led the entire tournament in Offensive VAEP (+5.965). He won the Golden Boot with 8 goals, including a hat-trick in the Final. His high volume of progressive dribbles, explosive take-ons into the penalty box, and high-conversion shooting generated unparalleled scoring-probability surges for France.
- Julián Álvarez:** Scored 4 open-play goals from only 293 total actions. His remarkable efficiency in converting low-probability transitions into goals (notably his solo run against Croatia) gave him the second-highest total VAEP in the tournament.
- Cody Gakpo:** Scored 3 group-stage goals with minimal attempts (left foot, right foot, and header), producing an extraordinarily high average value per action with almost no negative turnovers.
- Olivier Giroud:** Achieved +2.753 Total VAEP in just 151 actions. As a classic penalty-box target forward, he scored 4 crucial tournament goals with near-zero time wasted on non-dangerous possessions.
- Lionel Messi:** Messi recorded the second-highest Offensive VAEP (+5.119) of the World Cup, trailing only Mbappé. He was the tournament's primary playmaker (873 actions, 7 goals, 3 assists, Golden Ball winner). However, his Defensive VAEP was -2.391 — because Messi operates as the central attacking hub, missed through-balls and lost possessions in advanced areas mathematically elevate the opponent's counter-attack conceding probability ($P_{concedes}$), producing a net-negative defensive score that moderates his total combined rating relative to his offensive output.
- Bukayo Saka:** England's top offensive threat (3 goals). Saka demonstrated outstanding decision-making on the right flank, consistently completing progressive crosses and drawing key fouls inside the final third.
- Richarlison:** Scored 3 goals in 169 actions, highlighted by top-tier finishing — his acrobatic scissor/bicycle kick versus Serbia (visualized in the iconic-actions gallery above at +0.904) and his team goal versus South Korea.

8. **Gonalo Ramos:** The most efficient player per action in the tournament’s top 10. Ramos generated a massive +2.399 VAEP across only 84 total actions, driven by his hat-trick and assist against Switzerland in the Round of 16.
9. **Ritsu Doan:** Japan’s super-sub, who scored momentum-shifting equalizers against heavyweights Germany and Spain, generating high-leverage scoring-probability swings in very limited minutes.
10. **Jořko Gvardiol:** The standout defender: Gvardiol is the only pure defender to break into the Top 10, and, as the bar chart above shows, the only player in it with a positive Defensive VAEP. Across 1,033 actions, by far the highest action count on the list, he recorded a tournament-leading +0.811 Defensive VAEP alongside +1.487 Offensive VAEP. His performance was driven by goal-line clearances (vs. Belgium), penalty-box blocks, and line-breaking carries out of the defensive third.

Iconic FIFA World Cup 2022 Actions

Six of the tournament’s most recognizable moments were located on the pitch and re-visualized using their computed VAEP values, to connect the abstract probability model back to real, memorable football moments.

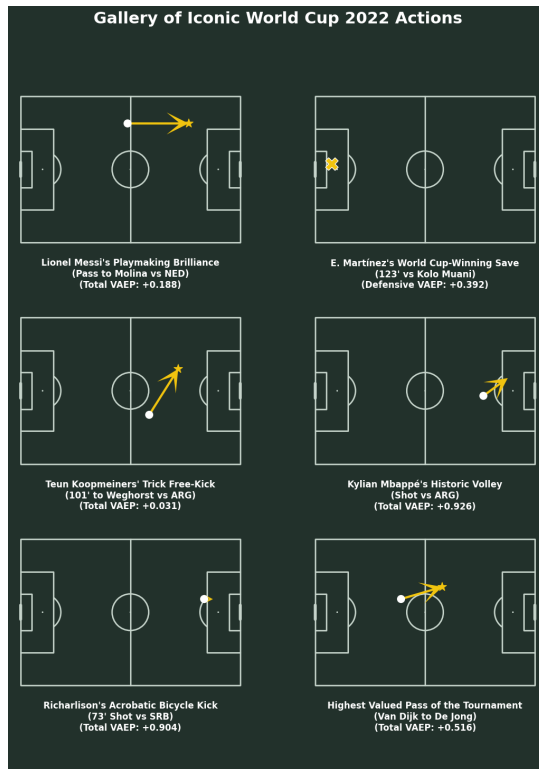


Fig. 7. Iconic FIFA World Cup 2022 actions and their VAEP values.

- **Lionel Messi’s pass to Molina (vs. Netherlands)** — Total VAEP +0.188. A mid-value progressive pass; meaningful, but far below a shot or save because a pass

alone rarely moves scoring probability as sharply as a shot on goal does.

- **Emiliano Martínez’s save at 123’ (vs. Kolo Muani, France)** — Defensive VAEP +0.392. One of the highest defensive values in the whole tournament, reflecting that this save directly prevented an almost-certain series-deciding goal in the dying seconds of extra time in the Final.
- **Teun Koopmeiners’ trick free-kick to Weghorst at 101’ (vs. Argentina)** — Total VAEP +0.031. A creative, low-percentage set piece that did not substantially raise scoring probability on its own — illustrating that VAEP rewards actions by their measured effect on outcome probability, not by their entertainment or novelty value.
- **Kylian Mbappé’s historic volley (vs. Argentina)** — Total VAEP +0.926. Near-maximal value, consistent with the top-actions table — a clinical finish converts scoring probability almost to certainty in one action.
- **Richarlison’s acrobatic bicycle kick at 73’ (vs. Serbia)** — Total VAEP +0.904. Also near-maximal, showing that VAEP does not discount a spectacular finish relative to a simple tap-in — the model only sees the change in scoring probability, which a converted shot maximizes regardless of technique.
- **Van Dijk’s pass to De Jong** — Total VAEP +0.516, flagged in this run as the single highest-valued pass of the tournament — a reminder that a well-timed progressive pass into a dangerous area can occasionally rival a shot in value, even though passes are worth much less than shots on average.

Together, the save (+0.392) and the two finishes (+0.926, +0.904) dominate this gallery numerically, while the free-kick attempt (+0.031) and the assist-creating pass (+0.188) sit much lower — a concrete, visual confirmation of the “goals/saves are rare-and-huge, passes are common-and-small” pattern described in Section 4’s action-type comparison.

6. Conclusion

The VAEP framework successfully transforms subjective match evaluation into an objective, probabilistic mathematical model. By leveraging SPADL’s standardized action language and 3-action game-state lookaheads ($k = 10$), every on-the-ball event across a full tournament is assigned a consistent, comparable valuation.

Across all five architectures tested — CatBoost, Calibrated LightGBM, XGBoost, Logistic Regression, and a Deep MLP, the three gradient-boosted tree ensembles (CatBoost, LightGBM, XGBoost) consistently outperformed both the linear baseline and the neural network on the unseen

2022 World Cup test set, clustering around 0.80–0.84 ROC-AUC with the lowest Brier score loss. XGBoost and Calibrated LightGBM were essentially tied for the best Scores-task discrimination (AUC 0.8202 each), while CatBoost held the best Concedes-task discrimination (AUC 0.8361). The Deep MLP was the weakest performer overall, particularly on the Concedes task (AUC 0.758), reinforcing that tree-based methods remain better suited than fully-connected neural networks to this kind of sparse, imbalanced, one-hot-heavy tabular event-stream data.

VAEP resolves the core blind spot of traditional Goals/Assists metrics. It objectively captures the high offensive contribution of world-class playmakers (Mbappé, Messi, Álvarez) while giving long-overdue credit to elite defensive contributions (Joško Gvardiol, Emiliano Martínez) — and, as the Traditional-Metrics-vs-VAEP comparison shows, it can also flag players like Julián Álvarez, Cody Gakpo, and Olivier Giroud as over-performing their box-score line, and players like Bruno Fernandes and Mislav Oršić as relying more heavily on a single decisive contribution than on sustained value creation across the match.